

LAB 1

Aim: To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

EXECUTION OF FOLLOWING CODE:

```

MySQL Shell 8.0.46

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Type '\help' or '? ' for help; '\quit' to exit.
MySQL JS > \sql
Switching to SQL mode... Commands end with ;
MySQL SQL > \connect root@localhost:3306
Creating a session to 'root@localhost:3306'
Please provide the password for 'root@localhost:3306': *****
Save password for 'root@localhost:3306'? [Y]es/[N]o/[e]ver (default No): N
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 12
Server version: 8.0.46 MySQL Community Server - GPL
No default schema selected; type \use <schema> to set one.
MySQL localhost:3306 ssl SQL >

```

SHOW databases;

CREATE DATABASE surabhi_09;

SHOW DATABASES;

```

MySQL localhost:3306 ssl SQL > SHOW databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.0028 sec)
MySQL localhost:3306 ssl SQL > CREATE DATABASE surabhi_09;
Query OK, 1 row affected (0.0091 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| surabhi_09 |
| sys |
+-----+
5 rows in set (0.0016 sec)

```

```
CREATE DATABASE stud1test;
```

```
SHOW DATABASES;
```

```
DROP DATABASE stud1test;
```

```
SHOW databases;
```

```
MySQL localhost:3306 ssl SQL > CREATE DATABASE surabhiltest;
Query OK, 1 row affected (0.0075 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| surabhiltest |
| surabhi_09 |
| sys |
+-----+
6 rows in set (0.0015 sec)
MySQL localhost:3306 ssl SQL > DROP DATABASE surabhiltest;
Query OK, 0 rows affected (0.0111 sec)
MySQL localhost:3306 ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| surabhi_09 |
| sys |
+-----+
5 rows in set (0.0014 sec)
```

```
USE surabhi_09;
```

```
CREATE TABLE Persons (
```

```
PersonID int,
```

```
LastName varchar(25),
```

```
FirstName varchar(25),
```

```
Address varchar(100),
```

```
City varchar(25)
```

```
);
```

```
SHOW TABLES;
```

```
DESCRIBE Persons;
```

```
DROP TABLE Persons;
```

```
MySQL localhost:3306 ssl surabhi_09 SQL > SHOW TABLES;
+-----+
| Tables_in_surabhi_09 |
+-----+
| persons               |
+-----+
1 row in set (0.0154 sec)
MySQL localhost:3306 ssl surabhi_09 SQL > DESCRIBE Persons;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| PersonID   | int           | YES  |     | NULL    |       |
| LastName   | varchar(25)   | YES  |     | NULL    |       |
| FirstName  | varchar(25)   | YES  |     | NULL    |       |
| Address    | varchar(100)  | YES  |     | NULL    |       |
| City       | varchar(25)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.0137 sec)
MySQL localhost:3306 ssl surabhi_09 SQL > DROP TABLE Persons;
Query OK, 0 rows affected (0.0903 sec)
```

CREATE TABLE Customers (CustomerID int, LastName varchar(25), FirstName
varchar(25), Address varchar(100), City varchar(25), primary key
(CustomerID));

DESCRIBE Customers;

INSERT INTO Customers (CustomerID, LastName, FirstName, Address, City)

VALUES (96365, 'Shah', 'Hiren', 'A67, Building10, abc lane', 'Mumbai');

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl surabhi_09 SQL > CREATE TABLE Customers (CustomerID int, LastName varchar(25), FirstName
-> varchar(25), Address varchar(100), City varchar(25), primary key
-> (CustomerID));
Query OK, 0 rows affected (0.1247 sec)
MySQL localhost:3306 ssl surabhi_09 SQL > DESCRIBE Customers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int           | NO   | PRI | NULL    |       |
| LastName   | varchar(25)   | YES  |     | NULL    |       |
| FirstName  | varchar(25)   | YES  |     | NULL    |       |
| Address    | varchar(100)  | YES  |     | NULL    |       |
| City       | varchar(25)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.0503 sec)
MySQL localhost:3306 ssl surabhi_09 SQL > INSERT INTO Customers (CustomerID, LastName, FirstName, Address, City)
-> VALUES (96365, 'Shah', 'Hiren', 'A67, Building10, abc lane', 'Mumbai');
Query OK, 1 row affected (0.0173 sec)
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT * FROM Customers;
+-----+-----+-----+-----+-----+
| CustomerID | LastName | FirstName | Address                | City |
+-----+-----+-----+-----+-----+
| 96365      | Shah    | Hiren    | A67, Building10, abc lane | Mumbai |
+-----+-----+-----+-----+-----+
1 row in set (0.0022 sec)
```

INSERT INTO Customers VALUES

(96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),

(89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');

----Now add approx. 10 customers-----

SELECT * FROM Customers;

```
MySQL localhost:3306 ssl surabhi_09 SQL > INSERT INTO Customers VALUES
-> (4, 'bansode', 'ekansh', 'vashi', 'navimumbai'),
-> (9, 'CHOPADEKAR', 'SURABHI', 'VIKROLI', 'MUMBAI'),
-> (13, 'DEVANG', 'PRATIK', 'SION', 'MUMBAI'),
-> (23, 'HANCHATE', 'TANAY', 'PAREL', 'MUMBAI'),
-> (31, 'KUKREJA', 'DRUSHTI', 'CHEMBUR', 'MUMBAI'),
-> (40, 'OAK', 'ANANNYA', 'THANE', 'THANE'),
-> (45, 'NAIR', 'TANISHRK', 'THANE', 'THANE'),
-> (58, 'Soni', 'kunal', 'Ambernath', 'Palghar'),
-> (59, 'SURVE', 'PARTH', 'BELAPUR', 'NAVIMUMBAI'),
-> (100, 'PERLA', 'AMOGH', 'BERHMPUR', 'IDK_CITY');

Query OK, 10 rows affected (0.0178 sec)

Records: 10 Duplicates: 0 Warnings: 0
MySQL localhost:3306 ssl surabhi_09 SQL >
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT * FROM Customers;
```

CustomerID	LastName	FirstName	Address	City
4	bansode	ekansh	vashi	navimumbai
9	CHOPADEKAR	SURABHI	VIKROLI	MUMBAI
13	DEVANG	PRATIK	SION	MUMBAI
23	HANCHATE	TANAY	PAREL	MUMBAI
31	KUKREJA	DRUSHTI	CHEMBUR	MUMBAI
40	OAK	ANANNYA	THANE	THANE
45	NAIR	TANISHRK	THANE	THANE
58	Soni	kunal	Ambernath	Palghar
59	SURVE	PARTH	BELAPUR	NAVIMUMBAI
100	PERLA	AMOGH	BERHMPUR	IDK_CITY
89790	Joshi	Ramesh	413,Sadashiv Nagar	Pune
96300	Singh	Prakesh	B7,House10,abc lane	Mumbai
96365	Shah	Hiren	A67,Building10,abc lane	Mumbai

```
13 rows in set (0.0046 sec)
```

ALTER TABLE Customers ADD Email varchar(50);

DESCRIBE Customers;

ALTER TABLE Customers ADD Remark varchar(255);

DESCRIBE Customers;

```
MySQL localhost:3306 ssl surabhi_09 SQL > ALTER TABLE Customers ADD Email varchar(50);
Query OK, 0 rows affected (0.0723 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:3306 ssl surabhi_09 SQL > DESCRIBE Customers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int | NO | PRI | NULL | |
| LastName | varchar(25) | YES | | NULL | |
| FirstName | varchar(25) | YES | | NULL | |
| Address | varchar(100) | YES | | NULL | |
| City | varchar(25) | YES | | NULL | |
| Email | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.0811 sec)

MySQL localhost:3306 ssl surabhi_09 SQL > ALTER TABLE Customers ADD Remark varchar(255);
Query OK, 0 rows affected (0.0183 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:3306 ssl surabhi_09 SQL > DESCRIBE Customers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int | NO | PRI | NULL | |
| LastName | varchar(25) | YES | | NULL | |
| FirstName | varchar(25) | YES | | NULL | |
| Address | varchar(100) | YES | | NULL | |
| City | varchar(25) | YES | | NULL | |
| Email | varchar(50) | YES | | NULL | |
| Remark | varchar(255) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.0026 sec)
```

SELECT * FROM Customers;

UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE
CustomerID=89790;

```
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT * FROM Customers;
+-----+-----+-----+-----+-----+-----+
| CustomerID | LastName | FirstName | Address | City | Email | Remark |
+-----+-----+-----+-----+-----+-----+
| 4 | bansode | ekansh | vashi | navimumbai | NULL | NULL |
| 9 | CHOPADEKAR | SURABHI | VIKROLI | MUMBAI | NULL | NULL |
| 13 | DEVANG | PRATIK | SION | MUMBAI | NULL | NULL |
| 23 | HANCHATE | TANAY | PAREL | MUMBAI | NULL | NULL |
| 31 | KUKREJA | DRUSHTI | CHEMBUR | MUMBAI | NULL | NULL |
| 40 | OAK | ANANNYA | THANE | THANE | NULL | NULL |
| 45 | NAIR | TANISHRK | THANE | THANE | NULL | NULL |
| 58 | Soni | kunal | Ambernath | Palghar | NULL | NULL |
| 59 | SURVE | PARTH | BELAPUR | NAVIMUMBAI | NULL | NULL |
| 100 | PERLA | AMOGH | BERHMPUR | IDK_CITY | NULL | NULL |
| 89790 | Joshi | Ramesh | 413,Sadashiv Nagar | Pune | NULL | NULL |
| 96300 | Singh | Prakesh | B7,House10,abc lane | Mumbai | NULL | NULL |
| 96365 | Shah | Hiren | A67,Building10,abc lane | Mumbai | NULL | NULL |
+-----+-----+-----+-----+-----+-----+
13 rows in set (0.0048 sec)

MySQL localhost:3306 ssl surabhi_09 SQL > UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE
-> CustomerID=89790;
Query OK, 1 row affected (0.0063 sec)

Rows matched: 1 Changed: 1 Warnings: 0
```

SELECT CustomerID,Address FROM Customers;

```
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT CustomerID,Address FROM Customers;
+-----+-----+
| CustomerID | Address |
+-----+-----+
| 4 | vashi |
| 9 | VIKROLI |
| 13 | SION |
| 23 | PAREL |
| 31 | CHEMBUR |
| 40 | THANE |
| 45 | THANE |
| 58 | Ambernath |
| 59 | BELAPUR |
| 100 | BERHMPUR |
| 89790 | 413,Sadashiv Nagar |
| 96300 | B7,House10,abc lane |
| 96365 | A67,Building10,abc lane |
+-----+-----+
13 rows in set (0.0020 sec)
```

-----Try yourself on different set of attributes-----

SELECT CustomerID,Address FROM Customers WHERE CustomerID=23;

```
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT CustomerID, Address FROM Customers WHERE CustomerID=23;
```

CustomerID	Address
23	PAREL

1 row in set (0.0017 sec)

SELECT * FROM Customers WHERE CustomerID=96300;

```
MySQL localhost:3306 ssl surabhi_09 SQL > SELECT * FROM Customers WHERE CustomerID=96300;
```

CustomerID	LastName	FirstName	Address	City	Email	Remark
96300	Singh	Prakesh	B7,House10,abc lane	Mumbai	NULL	NULL

1 row in set (0.0121 sec)

1. What is a database? Why do we need a database management system such as MySQL?

A database is an organized collection of related data that is stored electronically and can be easily accessed, managed, updated, and retrieved. It helps store large amounts of information systematically.

A Database Management System (DBMS) such as MySQL is software used to create, store, organize, retrieve, update, and manage this data efficiently. We need a DBMS to provide easy access to data, maintain its accuracy and consistency, reduce data redundancy, provide security, and handle large amounts of data efficiently. Thus, a database stores the data, while a DBMS such as MySQL provides the tools to manage and work with that data.

2. What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

The three SQL commands are used for different purposes:

- **CREATE DATABASE** – Used to create a new database in MySQL.
Example: CREATE DATABASE vedantitathe_62;
- **SHOW DATABASES** – Used to display a list of all databases available on the MySQL server.
Example: SHOW DATABASES;
- **DROP DATABASE** – Used to permanently delete an existing database along with all the tables and data stored in it.
Example: DROP DATABASE vedantitathe_62;

3. What is a table in a database? Explain the purpose of rows and columns in a table.

A table in a database is a structured collection of related data arranged in rows and columns. In the vedantitathe_62 database, the Customers table is used to store information about different customers.

Rows represent individual customer records in the table. For example, each row in the Customers table contains the details of one particular customer.

Columns represent the different attributes or fields of the customers. The Customers table contains columns such as CustomerID, LastName, FirstName, Address, City, Email, and Remark, with each column storing a specific type of information.

4. What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is an attribute that uniquely identifies each record in a table. The values of a primary key must be unique and cannot be NULL.

In the Customers table, CustomerID is selected as the primary key because it provides a unique identification for each customer. This prevents duplicate records from having the same identifier and helps maintain the accuracy and integrity of the data.

Thus, CustomerID serves as the unique identifier for each record in the Customers table.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

The VARCHAR and INT data types are used to define the type of data that can be stored in a column.

- VARCHAR is used to store variable-length character or text data. The maximum length is specified when defining the column.

Example: FirstName VARCHAR(50) can store a person's first name.

- INT is used to store whole numbers (integers), without decimal values.

Example: CustomerID INT can store a customer's identification number.

In the Customers table, VARCHAR is suitable for fields containing text, while INT is suitable for fields containing whole-number values.

6. What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

The INSERT, SELECT, UPDATE, and ALTER TABLE commands are used for different operations in a database:

- INSERT – Used to add new records or rows to a table.

Example: INSERT INTO Customers VALUES (...);

- **SELECT** – Used to retrieve or view data from a table.
Example: `SELECT * FROM Customers;`
- **UPDATE** – Used to modify existing data in one or more records.
Example: `UPDATE Customers SET City = 'Mumbai' WHERE CustomerID = 101;`
- **ALTER TABLE** – Used to modify the structure or definition of an existing table, such as adding, deleting, or modifying columns.
Example: `ALTER TABLE Customers ADD Phone VARCHAR(15);`

In short, INSERT adds data, SELECT retrieves data, UPDATE modifies existing data, and ALTER TABLE modifies the structure of a table

7. Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The WHERE clause is important in an UPDATE statement because it specifies which particular record or records should be modified. It ensures that only the intended data is changed.

If an UPDATE statement is written without a WHERE condition, the specified column will be updated for all records in the table. This may unintentionally modify data and cause loss of the original information.

For example, `UPDATE Customers SET City = 'Mumbai' WHERE CustomerID = 101;` will update only the record with the specified CustomerID. Without the WHERE clause, the City of every customer in the Customers table would be changed to Mumbai.